

Name:

Date:

Solve each problem. Show your work in the box.

Write your final answer on the answer line (or in the blanks shown).

Use the strategies you learned in Lesson 5.12.05.

1 Read a Straight Line Graph

A straight line graph shows apple cost: 1 kg = \$2, 2 kg = \$4, 3 kg = \$6.

How much do 5 kg of apples cost? Does this follow a straight line?

$\$2 \times 5 = \10 . Yes — the cost increases by \$2 for each kilogram, forming a straight line. ...

2 Identify the Rate from a Graph

A straight line graph shows distance vs. time: at 1 hour = 60 km, at 2 hours = 120 km, at 3 hours = 180 km.

What is the rate of travel in km/h? How far in 4.5 hours?

Rate = 60 km/h. Distance in 4.5 hours = $60 \times 4.5 = 270$ km. _____

3 Plot a Straight Line

A dog-walking service charges \$8 per hour.

Create a table for 1 to 5 hours, then describe the straight line graph it would produce.

1h=\$8, 2h=\$16, 3h=\$24, 4h=\$32, 5h=\$40. The graph is a straight line through the origin rising \$8 for each additional hour.

4 Interpret the Slope

A savings graph: Week 1 = \$15, Week 2 = \$30, Week 3 = \$45.

How much is saved each week? Write a sentence describing the pattern.

\$15 is saved each week. For every additional week, total savings increase by \$15.

5 Reasoning: Atlas vs. Mira

Atlas's plant growth graph is a straight line from (0, 0) to (6, 18). Mira says the plant grows 3 cm per day.

Is Mira correct? Find the growth rate and explain.

Rate = $18 \text{ cm} \div 6 \text{ days} = 3 \text{ cm/day}$. Yes, Mira is correct. _____